# Spectrally constrained stochastic factorization for low-dose CT

## 1 Low-dose CT reconstruction model

We reconstruct a two-dimensional attenuation image from X-ray measurements taken at several angles. Attenuation describes how strongly material reduces the transmitted intensity per unit distance. At one angular position, called a view, an X-ray source illuminates a line of detector bins. The source-to-bin rays spread out in a fan. Under the Beer–Lambert model, the negative logarithm of intensity relative to unobstructed illumination approximates a line integral of attenuation [2].

Treat $X_{pq}$ as the constant attenuation in pixel $(p, q)$ on a $512 \times 512$ grid. If $\ell_{ij,pq}$ is the length in millimeters of ray $j$ in view $i$ within that pixel, the calibrated forward map is

$$P_i \colon \mathbb{R}^{512\times 512} \longrightarrow \mathbb{R}^d, \qquad [P_i(X)]_j = \sum_{p,q} \ell_{ij,pq} X_{pq}, \qquad d = 956.$$

The fixed source and detector geometry determines these weights. ASTRA traces the rays to apply this map to selected views [4].

We write the image as $X = UV^\top$ with $U, V \in \mathbb{R}^{512\times 64}$, restricting its rank to at most 64. All views constrain the same factors. For measured view data $y_i \in \mathbb{R}^d$, the unweighted loss is

$$f(U, V) = \frac{1}{N} \sum_{i\in\mathcal{I}_{\text{train}}} \frac{\|P_i(UV^\top) - y_i\|_2^2}{2d}, \qquad N = |\mathcal{I}_{\text{train}}|.$$

The loss is quadratic in the image and nonconvex in the factors. To bound factor scale, use spectral-norm caps $C = \{(U, V) : \|U\|_{\text{op}} \le R_U,\ \|V\|_{\text{op}} \le R_V\}$. Each radius is twice the operator norm of the corresponding calibrated warm factor. Nonnegative factors give a nonnegative attenuation image, so define $D = \{(U, V) : U \ge 0,\ V \ge 0\}$ entrywise. Write $g = \iota_D$ for the function that is zero on $D$ and infinite elsewhere. Since $g$ acts directly on the factors, $T$ is the identity. With $x = (U, V)$, the problem has the SFRAMES form

$$\min_{x\in C}\{f(x) + g(Tx)\}.$$

The component gradients are Lipschitz on the compact region $C$. Strictly positive sufficiently small factors lie in the interiors of both $C$ and $D$.

The adjoint $P_i^*$, called backprojection, distributes each detector residual to the pixels crossed by its ray using the same ray-length weights. For a uniform minibatch $B$ of training views, define

$$E_B = \frac{1}{|B|d} \sum_{i\in B} P_i^*(P_i(UV^\top) - y_i), \qquad \widehat{\nabla} f_B(U, V) = (E_B V, E_B^\top U).$$

This estimator is unbiased conditional on the current factors. One joint gradient invocation applies one forward projection and one adjoint over $B$. An alternating method requests one of the two factor gradients at its block query point.

## 2 Data and evaluation protocol

The inputs are measured fan-beam sinograms from 2DeteCT [2]. A sinogram stacks the detector readings over the angular views of one slice. We fit the Mode 1 low-dose measurements. The image reference is the published Mode 2 reconstruction from a separate high-dose acquisition of the same slice. The two sample mixes label different specimen tubes, with three evaluation cross sections from each tube.

Each raw Mode 1 sinogram has 3,601 views and 1,912 detector cells. We discard the repeated final view and sum adjacent detector cells. A dark field measures detector offset and two flat fields measure unobstructed illumination. After the same detector binning, let $S_i, D_{\mathrm{dark}}$ and $F_{\mathrm{flat}}$ denote the measurement, dark field and mean flat field. The processed measurement before detector reversal is

$$-\log\left(\max\left\{\frac{S_i - D_{\mathrm{dark}}}{F_{\mathrm{flat}} - D_{\mathrm{dark}}}, 10^{-6}\right\}\right).$$

We reverse the detector axis to match the DeepInverse convention [3]. All eight selected acquisitions have zero transmission clipping counts. Each processed sinogram has $3{,}600 \times 956$ entries.

ASTRA 2.5.0 applies the CPU `line_fanflat` operator in millimeters [4]. The source-to-origin distance is 431.019989 mm, the origin-to-detector distance is 97.980499 mm and the binned detector spacing is 0.1496 mm. The square field has width 116.528143 mm and center $(-0.113797, -0.113797)$ mm. View $i$ has angle $\pi - 2\pi i/3600$. The factors use FP64 arithmetic and the projector uses FP32. To align the published $1024 \times 1024$ reference with the fitting grid, we flip it vertically, divide by its native voxel width 0.113797 mm, then average $2 \times 2$ pixel blocks.

Views have indices $0, \ldots, 3599$. The training set uses residue zero modulo ten for 360 views and residues zero, three, and six for 1,080 views. Both settings reserve residue nine for 360 held-out views. The 120 indices divisible by 30 calibrate the same warm start for both training sizes. Calibration views belong to both training sets. Held-out views are disjoint from training and calibration. Increasing the number of training views changes the finite sum while preserving the image and factor dimensions.

The development slices are 04300, 03150. Table 1 lists the evaluation slices. Each slice is reconstructed separately. The stochastic methods are paired at seed labels 202, 203, 204. Development uses oracle seed label 110. Each oracle generator uses its seed label plus 100,000. Calibration uses seed 110 in development and 201 in evaluation. It is fixed within a slice across methods, oracle seeds, and view counts. Full-batch SFRAMES and PALM each run once per slice and training size and provide shared references for the stochastic pairs.

Table 1: Fresh evaluation slices. All belong to the post-replacement detector regime.

| Sample mix | Evaluation slices |
|---|---|
| Mix 2 | 03041, 03381, 03611 |
| Mix 3 | 03791, 04131, 04931 |

## 3 Calibration and initialization

Let $P_J$ stack the 120 calibration views. We divide $(P_J^* y_J)_+$ entrywise by $P_J^* P_J \mathbf{1}$, floored at $10^{-3}$ times its maximum. From a thin SVD of this image, each rank-64 factor column uses the positive or negative singular-vector parts with the larger product of norms, with the corresponding singular-value scaling. We add independent uniform $[0, 1)$ entries multiplied by $10^{-3}$ times the RMS of each factor. Finally, we fit a scalar amplitude by least squares on the calibration views, with lower bound $10^{-6}$, and multiply both factors by its square root.

Write the resulting factors as $U_{\mathrm{cal}}, V_{\mathrm{cal}}$. The spectral radii are $R_U = 2\|U_{\mathrm{cal}}\|_{\mathrm{op}}$ and $R_V = 2\|V_{\mathrm{cal}}\|_{\mathrm{op}}$. All expanded evaluation runs start at these common feasible factors in $C \cap D$. The earlier LMO initialization outside $D$ belongs to the separate pilot ablation. The expanded trajectories therefore begin at $d_D = 0$, and the first SFRAMES update leaves $D$.

Starting with a normalized all-ones image, ten power iterations on $P_J^* P_J$ estimate its largest eigenvalue. Divide this estimate by $120d$ to obtain $\widehat{\lambda}$. The empirical curvature scales and smoothing prefactor are

$$L_U = \widehat{\lambda}\|V_{\mathrm{cal}}\|_{\mathrm{op}}^2, \qquad L_V = \widehat{\lambda}\|U_{\mathrm{cal}}\|_{\mathrm{op}}^2, \qquad \mu_0 = 1/\max(L_U, L_V).$$

These scales remain fixed during optimization.

Table 2: Calibrated spectral caps and empirical curvature scales. Values are rounded for display. Each slice shares these settings across methods, seeds, and training-view counts.

| Slice | $R_U$ | $R_V$ | $L_U$ | $L_V$ | $\mu_0$ |
|---|---|---|---|---|---|
| 04300 | 3.9529 | 3.9415 | 0.19332 | 0.19443 | 5.1431 |
| 03150 | 3.7934 | 3.7924 | 0.17897 | 0.17906 | 5.5846 |
| 03041 | 3.7261 | 3.7247 | 0.17264 | 0.17277 | 5.7881 |
| 03381 | 3.4379 | 3.435 | 0.14682 | 0.14707 | 6.7994 |
| 03611 | 3.8419 | 3.8475 | 0.18421 | 0.18367 | 5.4286 |
| 03791 | 3.7472 | 3.7484 | 0.17484 | 0.17472 | 5.7195 |
| 04131 | 3.7834 | 3.7811 | 0.1779 | 0.17812 | 5.6141 |
| 04931 | 3.9813 | 3.9806 | 0.19718 | 0.19724 | 5.07 |

# 4 Optimizers and rate selection

For $k \geq 0$, both SFRAMES variants use

$$\gamma_k = \gamma_0/(k+1)^{3/4}, \qquad \rho_k = 1/(k+1)^{1/2}, \qquad \mu_k = \mu_0/(k+1)^{1/4}.$$

Minibatch SFRAMES draws 64 training views uniformly without replacement at each step, independently across steps. Its full-batch counterpart uses every training view and tunes its own $\gamma_0$. The momentum starts with the first gradient and subsequently satisfies

$$m_k = (1-\rho_k)m_{k-1} + \rho_k \widehat{\nabla} f_{B_k}(x_k), \qquad k \geq 1.$$

Both add the orthant Moreau gradient after this averaging and update by

$$s_k = \mathrm{LMO}_C\left(m_k + \frac{x_k - \Pi_D(x_k)}{\mu_k}\right), \qquad x_{k+1} = (1-\gamma_k)x_k + \gamma_k s_k.$$

Here $\Pi_D$ clips negative entries, so the Moreau term is the negative part of each factor divided by $\mu_k$ [1]. For a block direction with thin singular value decomposition $G = A\,\mathrm{diag}(\sigma)B^\top$, the exact oracle is $-RA_+B_+^\top$, with singular vectors restricted to positive singular values. Its optimal value is $-R\|G\|_*$, where the star denotes the nuclear norm. The FP64 implementation retains singular values exceeding $\varepsilon_{\mathrm{mach}} \max(\mathrm{shape}\, G)\sigma_{\max}(G)$ and returns zero for a zero direction. Discarding small singular values gives an oracle support-gap bound of $R$ times their sum. Each SFRAMES update uses two $512 \times 64$ thin SVDs, one for each factor.

SPRING–SARAH updates $U$, then $V$ using the new $U$, with steps $a/L_U$ and $a/L_V$. Each block independently refreshes its estimator with probability $1/q$ using all training views. Both blocks begin with full gradients at their respective query points. Between refreshes, a 64-view SARAH difference updates each estimator [5]. ProxSARAH uses a joint step $a/(L_U + L_V)$, a 256-view snapshot, then $m$ recursive updates with 64-view differences before the next snapshot. Its relaxation is 0.8 [6]. Each recursive difference evaluates the same sampled views at its two query points. All subset samples are uniform without replacement within a batch. PALM uses every training view per alternating block step with the same curvature scaling as SPRING–SARAH [7]. Proximal block steps project onto the spectral-ball and orthant intersection using a single-set shortcut when feasible and Dykstra iterations with a numerical certificate otherwise. The individual projections clip negative entries or clip singular values at the radius. Dykstra uses a requested distance tolerance $10^{-8}/(k+1)^{1.1}$, at most 2,000 inner iterations, and certificate checks after the first iteration, every five iterations, and at the final iteration. The certificate uses a primal–dual gap with a roundoff allowance. A reported floating-point floor can terminate the projection before the requested tolerance is attained. The work count includes both evaluations in each recursive difference and all views used for refreshes and snapshots.

Table 3: Development grid and frozen evaluation settings. Pairs list the step multiplier followed by the refresh period for SPRING–SARAH or the inner-loop length for ProxSARAH.

| Variant | Candidates | Selected |
|---|---|---|
| Minibatch SFRAMES | 0.05, 0.1, 0.16, 0.25 | 0.1 |
| Full-batch SFRAMES | 0.1, 0.25, 0.5, 0.8 | 0.1 |
| SPRING–SARAH | (0.7, 16), (1.1, 16), (1.6, 16), (1.1, 8), (1.1, 32) | (1.6, 16) |
| ProxSARAH | (0.9, 12), (1.5, 12), (2.3, 12), (1.5, 6), (1.5, 24) | (2.3, 12) |
| PALM | 0.6, 1.1, 1.6, 2.3 | 2.3 |

We select each setting by the smallest median ratio of repaired training loss to initial loss across both development slices and both training sizes. The ratio uses the last iterate within 8,640 forward and adjoint views in each direction. This gives 88 development runs across two slices, two training sizes and 22 candidate settings. Held-out loss and reference-image error are reporting metrics. The selected rate lies at the upper grid edge for SPRING–SARAH, ProxSARAH, PALM. The selected rate lies at the lower grid edge for Full-batch SFRAMES. The comparison reflects the settings covered by these grids.

# 5 Evaluation results

132 of 132 evaluation runs completed. The comparison uses the last saved iterate with at most 17,280 forward views and 17,280 adjoint views. A run stops after the first complete update that reaches the work budget. When an update crosses the budget, the previous iterate supplies the within-budget comparison. The tables use saved probes without interpolation.

An iteration completes one update of both factors. A gradient call applies one forward operator and one adjoint to the requested batch and returns either a joint gradient or one block gradient. Both evaluations in each SARAH difference and all refresh and snapshot views contribute to the work count. Gradient-call counts therefore describe the implementation, while the two view counts measure how much projection data the calls process.

Raw loss uses the actual factors. Repaired loss, held-out loss and reference error use the image formed after taking each factor's positive part and scaling it down if its operator norm exceeds its cap. Spectral violation is $d_C = \max\{(\|U\|_{\mathrm{op}} - R_U)_+, (\|V\|_{\mathrm{op}} - R_V)_+\}$. Distance to $D$ is $d_D = (\| \min(U, 0)\|_F^2 + \| \min(V, 0)\|_F^2)^{1/2}$. Reference error is the Frobenius error relative to the aligned Mode 2 image, divided by that image's Frobenius norm.

For absolute metrics we take the median over seeds within each slice and then the median across slices at each training size. Percent differences first compare paired runs using the comparator in the denominator and then apply the same aggregation. Each training size has 18 paired seed cases from six slices. Full-batch SFRAMES and PALM provide shared deterministic references. These summaries are descriptive because the three slices from each sample mix share a sample tube.

Table 4: Work at the projection budget of 17,280 views per direction. Values are medians across slice medians. The last saved iterate can fall below the cap when the next complete update would exceed it.

| $N$ | Variant | Iterations | Gradient calls | Forward views | Adjoint views |
|---|---|---|---|---|---|
| 360 | Minibatch SFRAMES | 270 | 270 | 17280 | 17280 |
| 360 | Full-batch SFRAMES | 48 | 48 | 17280 | 17280 |
| 360 | SPRING–SARAH | 60 | 233 | 17144 | 17144 |
| 360 | ProxSARAH | 125 | 240 | 17280 | 17280 |
| 360 | PALM | 24 | 48 | 17280 | 17280 |
| 1,080 | Minibatch SFRAMES | 270 | 270 | 17280 | 17280 |
| 1,080 | Full-batch SFRAMES | 16 | 16 | 17280 | 17280 |
| 1,080 | SPRING–SARAH | 47 | 183 | 16792 | 16792 |
| 1,080 | ProxSARAH | 125 | 240 | 17280 | 17280 |
| 1,080 | PALM | 8 | 16 | 17280 | 17280 |

Table 5: Losses, image error and feasibility at the saved within-budget iterates in Table 4. All losses use the same detector and view normalization. Values are medians across slice medians.

| $N$ | Variant | Raw | Repaired | Held-out | Ref. error | $d_C$ | $d_D$ |
|---|---|---|---|---|---|---|---|
| 360 | Minibatch SFRAMES | 0.01219 | 0.0122 | 0.01238 | 0.3636 | 0 | 0.0104 |
| 360 | Full-batch SFRAMES | 0.0174 | 0.01746 | 0.0175 | 0.4983 | 0 | 0.0304 |
| 360 | SPRING–SARAH | 0.01613 | 0.01613 | 0.01615 | 0.4904 | 0 | 0 |
| 360 | ProxSARAH | 0.01543 | 0.01543 | 0.01547 | 0.4751 | 0 | 0 |
| 360 | PALM | 0.01909 | 0.01909 | 0.01909 | 0.5397 | 0 | 0 |
| 1,080 | Minibatch SFRAMES | 0.01225 | 0.01226 | 0.01235 | 0.3572 | 0 | 0.0102 |
| 1,080 | Full-batch SFRAMES | 0.02292 | 0.02304 | 0.02299 | 0.6154 | 0 | 0.0661 |
| 1,080 | SPRING–SARAH | 0.01735 | 0.01735 | 0.01734 | 0.5116 | 0 | 0 |
| 1,080 | ProxSARAH | 0.01543 | 0.01543 | 0.01543 | 0.4737 | 0 | 0 |
| 1,080 | PALM | 0.02362 | 0.02362 | 0.02358 | 0.6165 | 0 | 0 |

Table 6: Median paired percent differences for minibatch SFRAMES at 17,280 views per direction. Negative values favor minibatch SFRAMES. Slices won counts slices with lower median paired repaired loss.

| $N$ | Comparator | Pairs | Repaired | Held-out | Slices won |
|---|---|---|---|---|---|
| 360 | Full-batch SFRAMES | 18 | −30.84% | −29.65% | 6/6 |
| 360 | SPRING–SARAH | 18 | −24.55% | −23.55% | 6/6 |
| 360 | ProxSARAH | 18 | −21.48% | −20.52% | 6/6 |
| 360 | PALM | 18 | −36.63% | −35.62% | 6/6 |
| 1,080 | Full-batch SFRAMES | 18 | −48.70% | −48.37% | 6/6 |
| 1,080 | SPRING–SARAH | 18 | −29.39% | −28.93% | 6/6 |
| 1,080 | ProxSARAH | 18 | −21.07% | −20.72% | 6/6 |
| 1,080 | PALM | 18 | −51.31% | −50.99% | 6/6 |

At the common work cap, minibatch SFRAMES completes 270 updates at both training sizes. Full-batch SFRAMES completes 48 updates with 360 training views and 16 with 1,080 views. The median paired reduction in repaired training loss relative to full-batch SFRAMES is 30.8% with 360 training views and 48.7% with 1,080 training views. Minibatch SFRAMES has lower median paired repaired loss than each comparator on all six evaluation slices at both training sizes. Its held-out loss also improves relative to every comparator in the overall slice summaries. Figure 3 shows the individual slice effects so that the variability across slices remains visible.

Figures 1 and 2 show the saved loss and feasibility trajectories. These expanded runs use the shared feasible warm start described above. Both SFRAMES variants leave $D$ after the first update and subsequently reduce their orthant distance. Their loss after feasibility repair remains close to their raw loss at the work cap. Every saved evaluation probe has zero spectral violation, and the projection baselines retain nonnegative factors. At every evaluation baseline projection, clipping to the orthant already satisfies the spectral cap, so the single-set shortcut applies. The separately tuned rates and the boundary selections in the development grid limit the comparison to these practical settings.

# 6 Optimization trajectories

Figures 1 and 2 show all five methods at both training-view counts. For display, each saved seed trajectory is linearly interpolated between diagnostic probes on a common work grid. We take the pointwise median over seeds within each slice, then the median across the six slices. Full-batch SFRAMES and PALM contribute one run per slice and training size. Shading shows the middle 50% of slice medians for the two SFRAMES variants. These are descriptive trajectories, with no extrapolation beyond recorded work. Numerical tables use actual saved within-budget iterates, so a displayed interpolated endpoint can differ slightly from a table entry.

Probes were scheduled every 4,320 forward views in evaluation and every 2,160 in development, with extra probes at the first update and around budget crossings. The spacing limits what can be inferred about behavior between probes. Forward and adjoint view counts agree at every saved probe.

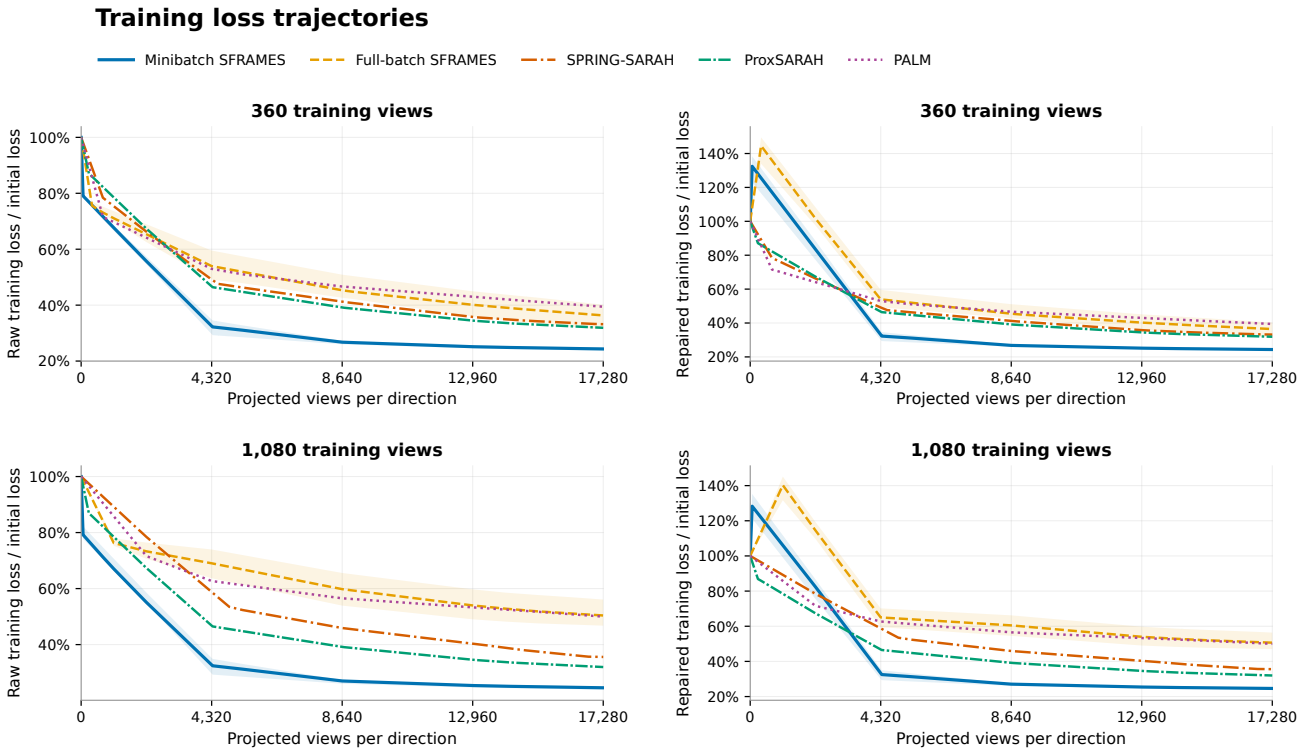



Figure 1: Raw and repaired training loss versus processed views in each projection direction. Each loss is divided by its own slice and training split's common initial loss before aggregation. Rows compare 360 and 1,080 training views.

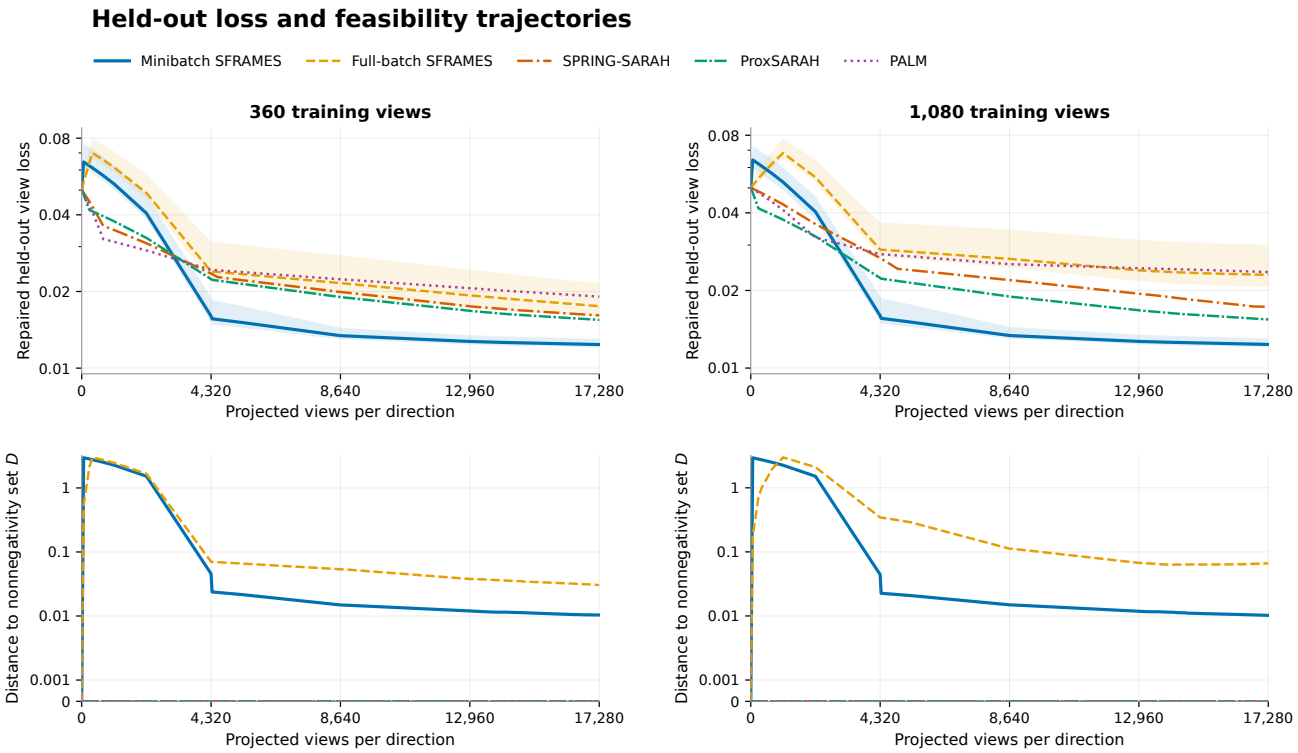



Figure 2: Held-out loss after factor repair and distance of the raw factors to the nonnegative orthant $D$, versus projection work. The feasibility axis uses a symmetric logarithmic scale with a linear region around zero. The shared warm start has zero distance to $D$. SFRAMES leaves $D$ after its first update. The three projection baselines coincide at zero distance throughout.

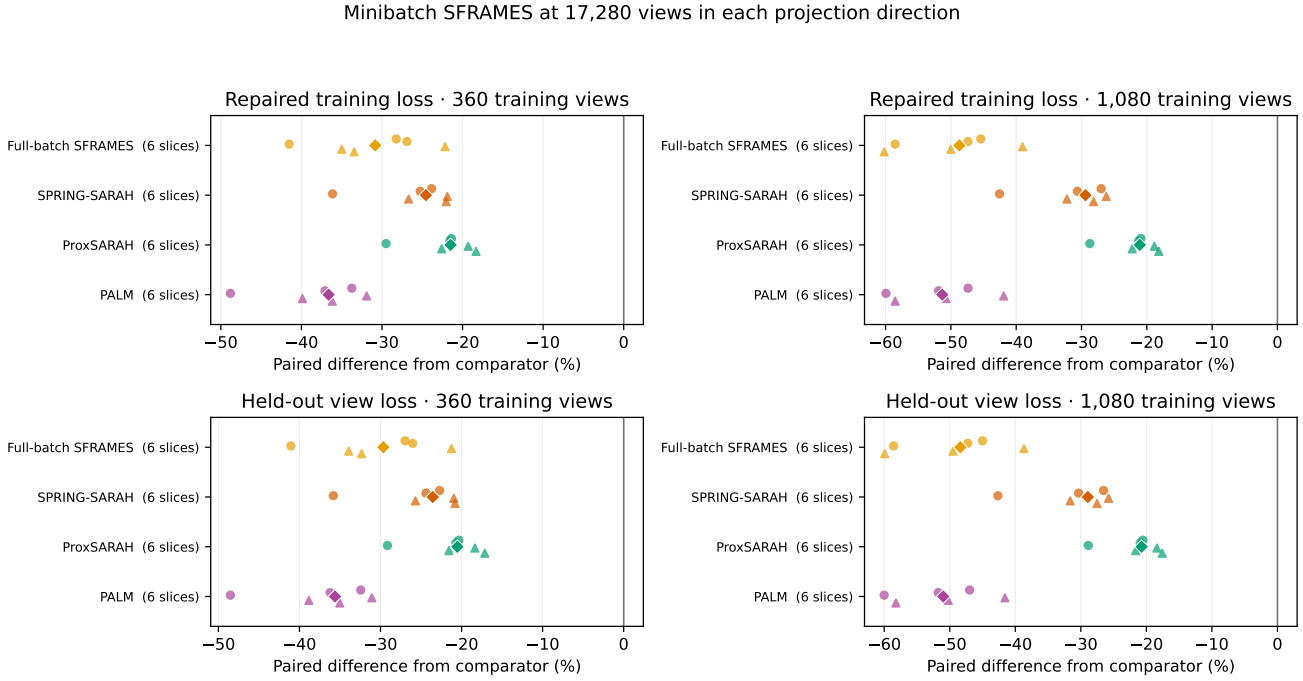



Figure 3: Paired effects for minibatch SFRAMES at the saved within-budget iterates. Small marks give the median paired effect over seeds for each slice. Circles and triangles distinguish the two sample mixes, and diamonds give the median across slices. Negative values favor minibatch SFRAMES.

## 7 Reproduction and scope

The expanded campaign contains 88 development runs and 132 fresh evaluation runs. Per-run configurations and original source, data and calibration hashes are stored under `runs/`. The files `manifest.json`, `selection.json` and `calibration.json` record the design, frozen rate choices and calibration checksums. Data sidecars `data/slice*.json` record the Zenodo sources, preprocessing conventions and original TIFF checksums. `reports/summary.json` records work-based endpoint and paired summaries. `analyze.py`, `plot_trajectories.py` and `report_latex.py` regenerate the analysis, figures and this document from saved results.

The comparisons measure optimization progress under the recorded operator work budget. Concurrent execution on the host does not provide a controlled timing benchmark. This report uses iterations, gradient-oracle calls and projection work to characterize the completed runs. The problem size, rank, initialization, smoothing schedule and projection settings are fixed in this campaign. Training-view count, stochastic seed and the development rate and refresh grids supply the reported variations.

## References


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